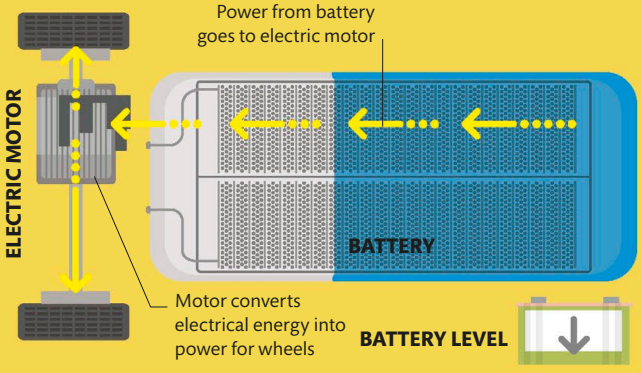
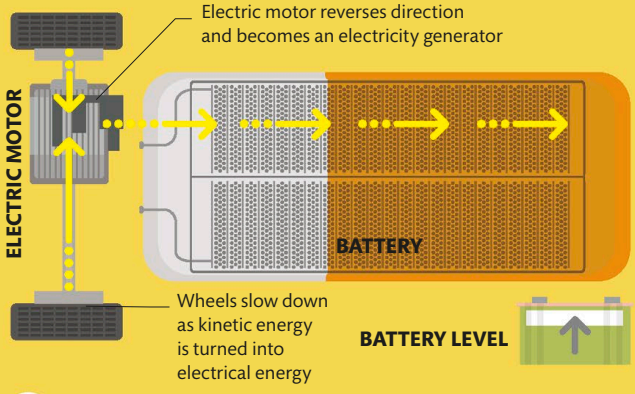
Regenerative braking
Most cars brake by using friction pads (see p.45), which transform the wheels' kinetic energy into wasted heat energy. Electric and hybrid cars convert the wheels' energy into electrical energy instead to recharge the battery.

THE FIRST ELECTRIC CAR WAS BUILT BY INVENTOR ROBERT ANDERSON IN THE 1830s

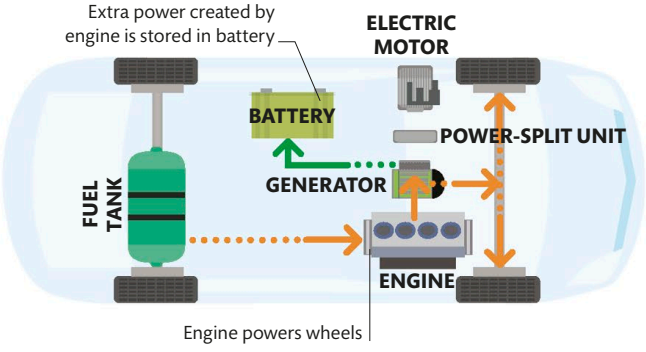




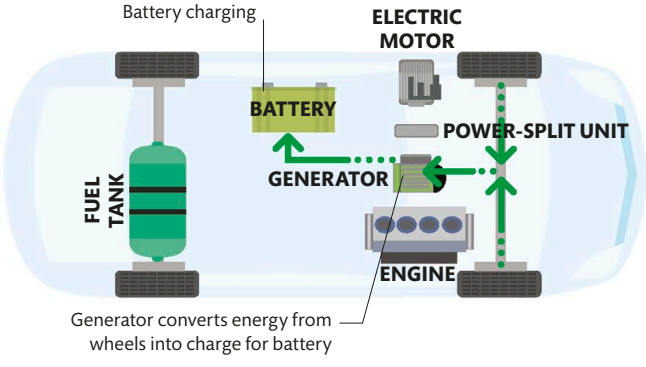
1 Acceleration
When an electric or hybrid car accelerates, its motor draws the energy it needs from the battery. The motor converts the battery's electrical energy into the car's kinetic energy. The level of charge in the battery falls as it is progressively drained of energy.

2 Braking
When the driver applies the brakes, the electric motor becomes a generator. Instead of drawing energy from the battery, it transforms the kinetic energy of the car's spinning wheels into electrical energy, which is returned to the battery to be reused.

Cruising
While the car is cruising at high speed on a long trip, the internal combustion engine operates on its own. The electric motor is not needed.



Braking
When the car begins to slow down, the internal combustion engine and electric motor switch off. During braking, the car's excess energy is converted into electricity to charge the battery.



DRIVERLESS CARS

A driverless car has various cameras, lasers, and radar that create a real-time 3-D image of the car's surroundings. Together with computers, satellite navigation, and artificial intelligence (AI), these enable the car to drive itself.

